

Session 5

1. list the 7 main phases of the Software Testing Life Cycle (STLC) and write one line describing what happens in each phase for a Login feature like the one in Instagram or Paytm.

1. Seven Main Phases of the Software Testing Life Cycle (STLC)

The **Software Testing Life Cycle (STLC)** is a sequence of activities performed by the testing team to ensure that a software feature works correctly.

Sr. No.	STLC Phase	Description for a Login Feature
1	Requirement Analysis	The tester studies the login requirements, such as valid credentials, invalid passwords, OTP, and forgot-password functionality.
2	Test Planning	The Test Lead decides the testing scope, resources, tools, schedule, and types of testing required for Login.
3	Test Case Design / Test Case Development	Testers prepare test cases for valid login, invalid login, empty fields, password rules, and other scenarios.
4	Test Environment Setup	The team prepares the required application build, database, devices, browsers, and test accounts.
5	Test Execution	Testers execute the test cases and compare the actual results with the expected results.
6	Defect Reporting / Test Cycle Closure	Defects are reported, fixed, and retested; after testing is completed, the team prepares the final test report and closes the cycle.

Note: Many STLC models describe **7 phases** by separating **Defect Reporting** and **Test Cycle Closure**. The six rows above combine the last two, so the seven-phase version is shown below:

Correct 7-Phase STLC List

1. Requirement Analysis

2. Test Planning
3. Test Case Design / Development
4. Test Environment Setup
5. Test Execution
6. Defect Reporting
7. Test Cycle Closure

2.

For a Flipkart-style Login module, create a table that lists the entry and exit criteria for each STLC phase (Planning, Monitoring, Analysis, Design, Implementation, Execution, Reporting, Closure).

2. Entry and Exit Criteria for Each STLC Phase

Project: Flipkart-style Login Module

Entry Criteria: Conditions that must be satisfied before a phase starts.

Exit Criteria: Conditions that must be satisfied before a phase is completed.

STLC Phase	Entry Criteria	Exit Criteria
1. Planning	Login requirements and project details are available.	Test plan is approved, scope is defined, resources are assigned, and schedule is finalized.
2. Monitoring	Test planning has started and testing activities are defined.	Testing progress, effort, defects, and risks are monitored and status updates are shared.
3. Analysis	Approved requirements and business rules are available.	Test scenarios are identified, requirements are reviewed, and test conditions are finalized.
4. Design	Test scenarios and requirements are available.	Test cases, test data, and expected results are prepared and reviewed.
5. Implementation	Test cases and test data are ready.	Test environment is configured, test cases are organized, and required accounts/builds are available.
6. Execution	Application build and test environment are ready; test cases are approved.	Test cases are executed, results are recorded, and defects are reported and retested.

7. Reporting	Test execution results and defect details are available.	Test Summary Report is prepared with pass/fail counts, defect status, and testing observations.
8. Closure	Testing objectives are completed and closure information is available.	Final report is approved, lessons learned are documented, and testing is formally closed.

Note: Monitoring is generally a continuous activity throughout STLC, while Reporting and Closure are often treated as separate activities.

**3 . Write a sample Test Plan section (max 1 page) for the Login module of a Zomato-like app, including objectives, scope, resources, and schedule.

Hint: Focus on what will be tested, who will test, and the timeline for testing the Login feature.**

3. Sample Test Plan — Login Module of a Zomato-like App

Test Plan Title: Login Module Test Plan
Application: Zomato-like Food Delivery Application
Feature: User Login
Prepared By: QA Testing Team
Version: 1.0

1. Objectives

- Verify that users can log in using valid credentials.
- Ensure that invalid credentials are rejected with appropriate error messages.
- Check validation for empty fields and incorrect input formats.
- Verify password visibility, Forgot Password, and account security-related functionality.
- Ensure that the Login feature works correctly across supported browsers and devices.

2. Scope

In Scope:

- Login using registered mobile number or email.
- Password and OTP validation, if supported.
- Valid and invalid login attempts.

- Empty-field and input-format validation.
- Forgot Password functionality.
- Login button, error messages, and navigation after successful login.
- Basic compatibility and usability testing.

Out of Scope:

- Restaurant search, food ordering, payment, delivery tracking, and other modules.
- Third-party services that are not part of the Login feature, except where required for login verification.

3. Resources and Responsibilities

Resource	Responsibility
Test Lead	Prepare the test plan, assign tasks, monitor progress, and review reports.
Tester / QA Engineer	Design test cases, execute tests, report defects, and perform retesting.
Developer	Fix reported defects and provide updated builds.
Product Owner	Clarify requirements and approve business expectations.
Test Environment	Web/mobile application, supported browsers, test accounts, and test database.
Tools	Jira or similar defect-tracking tool, browser DevTools, and test-case management tool.

4. Schedule

Activity	Duration
Requirement Analysis	Day 1
Test Planning and Test Case Design	Day 2
Test Environment Setup	Day 3
Test Execution	Days 4–5
Defect Retesting and Regression Testing	Day 6
Test Summary Report and Closure	Day 7

Deliverables: Test Plan, Test Cases, Defect Reports, Test Execution Results, and Test Summary Report.

4. Design 3 test cases for the Login module of a BookMyShow-style app, covering positive, negative, and boundary scenarios. Include test steps, expected results, and actual results columns.

Test Cases for BookMyShow-style Login Module

Assumption: The Login module supports a registered mobile number and password. The boundary scenario checks the minimum and maximum allowed password length according to the application's requirements.

Test Case ID	Scenario	Test Steps	Expected Result	Actual Result
TC_LOGIN_01	Positive: Login with valid credentials	1. Open the application. 2. Click Login . 3. Enter a registered mobile number/email. 4. Enter the correct password. 5. Click Login .	User should log in successfully and be redirected to the home page or account dashboard.	Pass: User logged in successfully and was redirected to the home page.
TC_LOGIN_02	Negative: Login with an incorrect password	1. Open the Login page. 2. Enter a registered mobile number/email. 3. Enter an incorrect password. 4. Click Login .	Login should be rejected, and a clear error message such as " Invalid credentials " should be displayed.	Pass: Login was rejected and an appropriate error message was displayed.

TC_LOGIN_03	Boundary: Password at minimum and maximum allowed length	1. Open the Login page. 2. Enter a valid registered mobile number/email. 3. Enter a password at the minimum allowed length and attempt login. 4. Repeat using a password at the maximum allowed length. 5. Record the results.	The application should accept input within the allowed length limits and reject passwords that exceed the maximum limit, showing proper validation.	Pass: Boundary-length inputs were handled according to the defined validation rules.
-------------	--	--	---	--

Important: The actual results above are **sample assumed results** for an assignment. In real testing, actual results must be recorded only after executing the application.

**5. After executing your test cases, write a short Test Summary Report for the Login module, mentioning how many tests passed, failed, and any major issues found.

Hint: Imagine you are sending this summary to your team lead after finishing Login testing.**

Test Summary Report — Login Module

Test Summary Report

Project: BookMyShow-style Application
Module: Login
Testing Type: Functional Testing
Test Execution Date: 16 September 2026
Prepared By: QA Testing Team

Execution Summary

Metric	Result
--------	--------

Total Test Cases	3
Passed	3
Failed	0
Blocked	0
Pass Percentage	100%

Major Issues Found

- No major issues were found during the assumed test execution.
- Valid credentials allowed successful login.
- Invalid credentials were rejected with an appropriate error message.
- Boundary input validation worked according to the defined requirements.